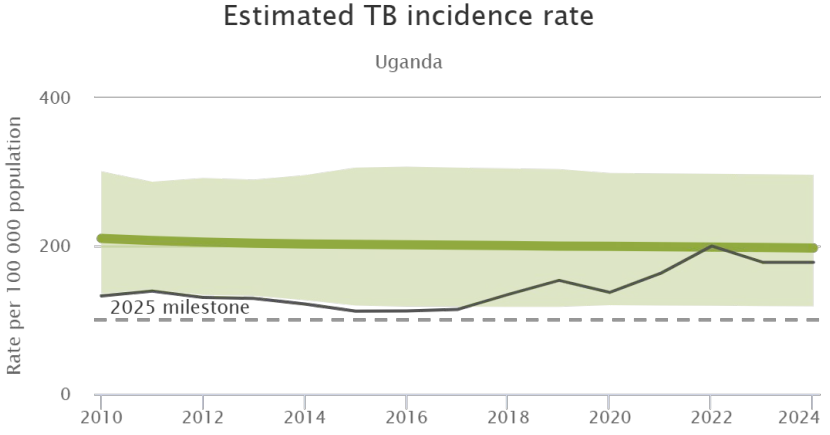
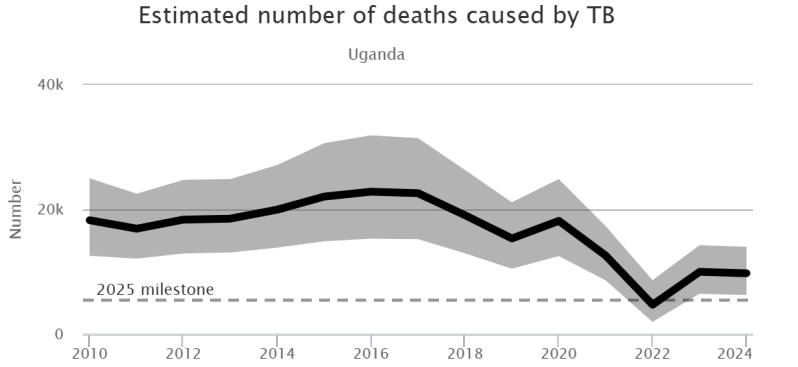
Bioreactor

By Group CSEEE-21

Context:

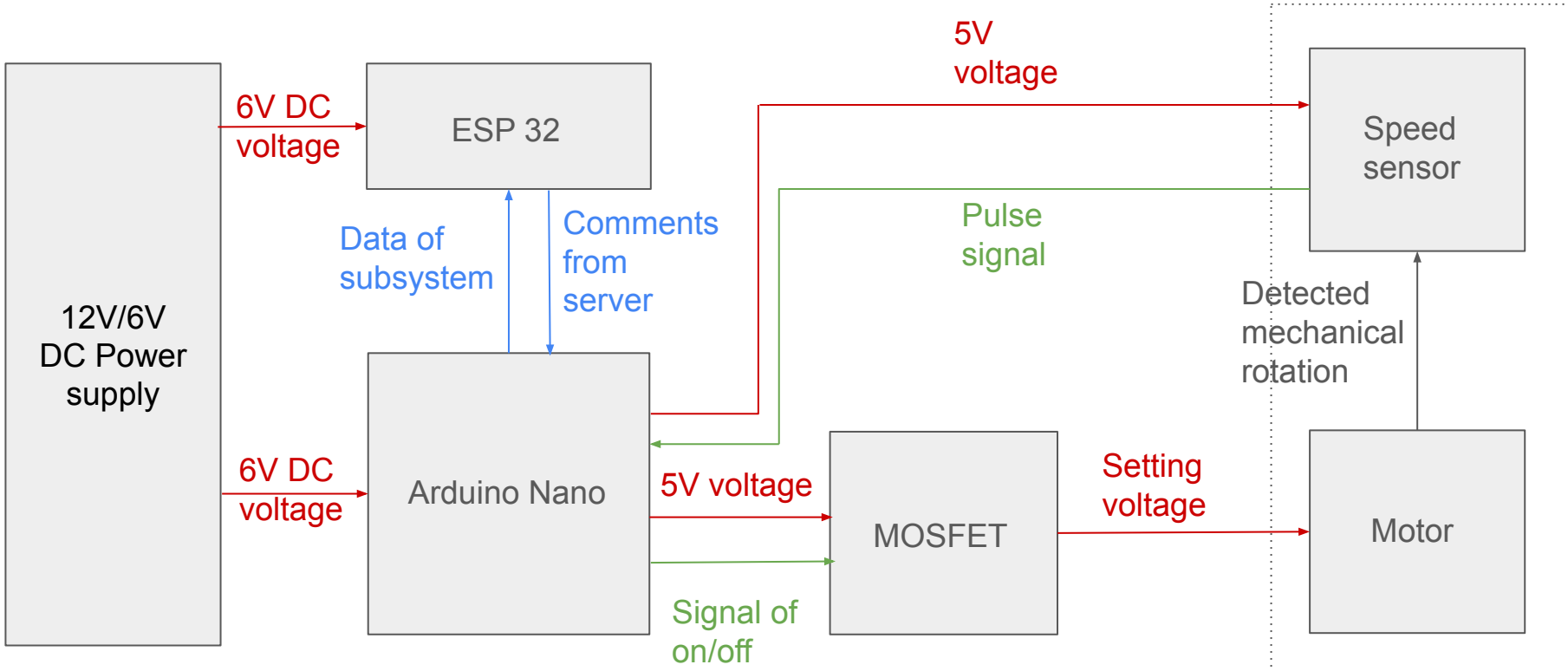
- Tuberculosis(TB) is a bacterial infection usually affecting the lungs
- Most common treatment of TB is through special antibiotics, such as rifampicin or isoniazid etc.
- Only effective if the TB they have contracted is not Multi-drug resistant(MDR).
- “Only about 2 in 5 people with multidrug-resistant TB accessed treatment in 2024.”, according to the WHO.
- Prevention may also be an option
- A vaccine trains the immune system to attack the actual bacterium so isn't affected by the various strains
- Often cheaper as it is a one-time dose compared to the months of treatment on antibiotic



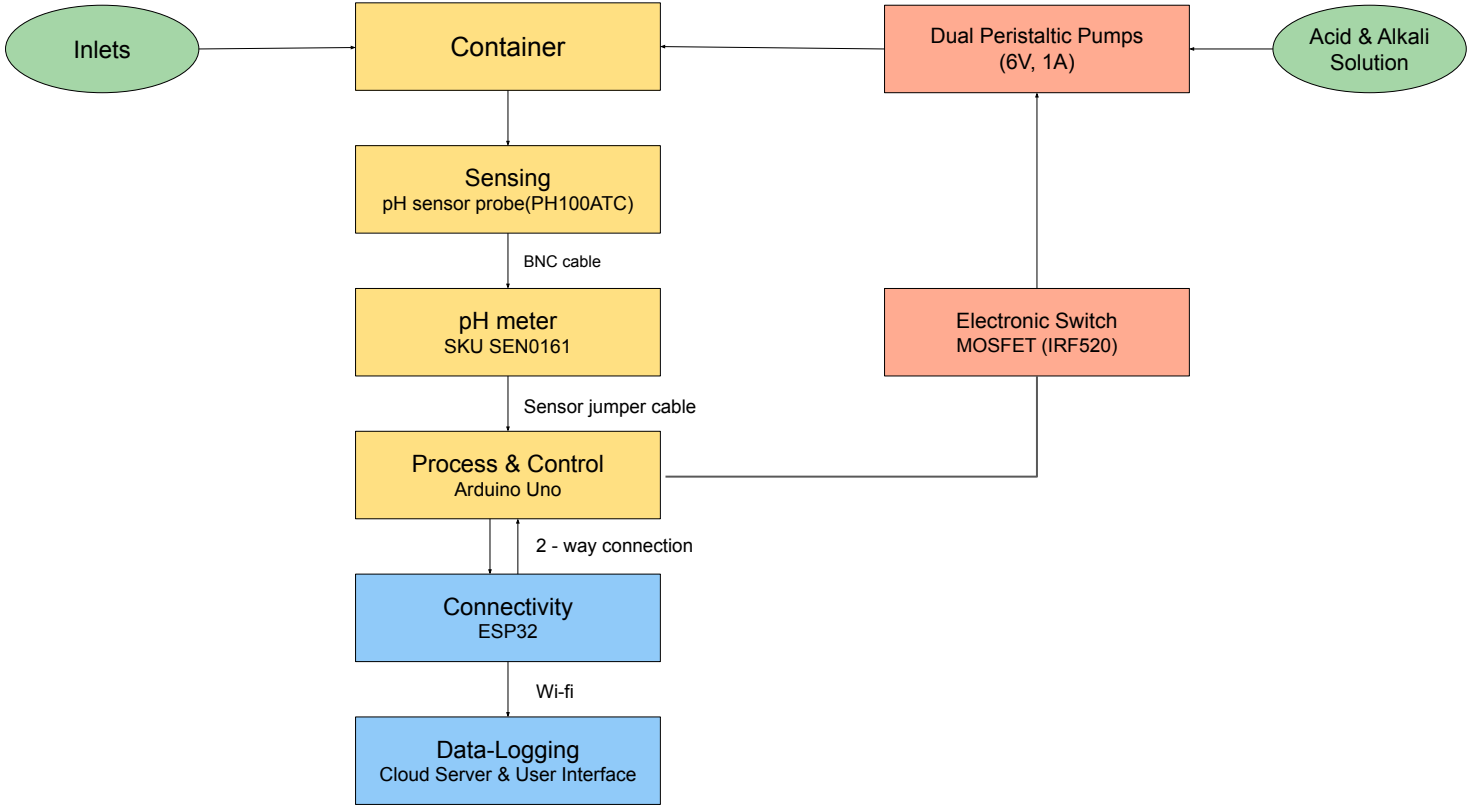


Source:https://worldhealthorg.shinyapps.io/tb_profiles/?_inputs_&tab=%22charts%22&lan=%22EN%22&iso3=%22UGA%22&entity_type=%22country%22

Stirring subsystem

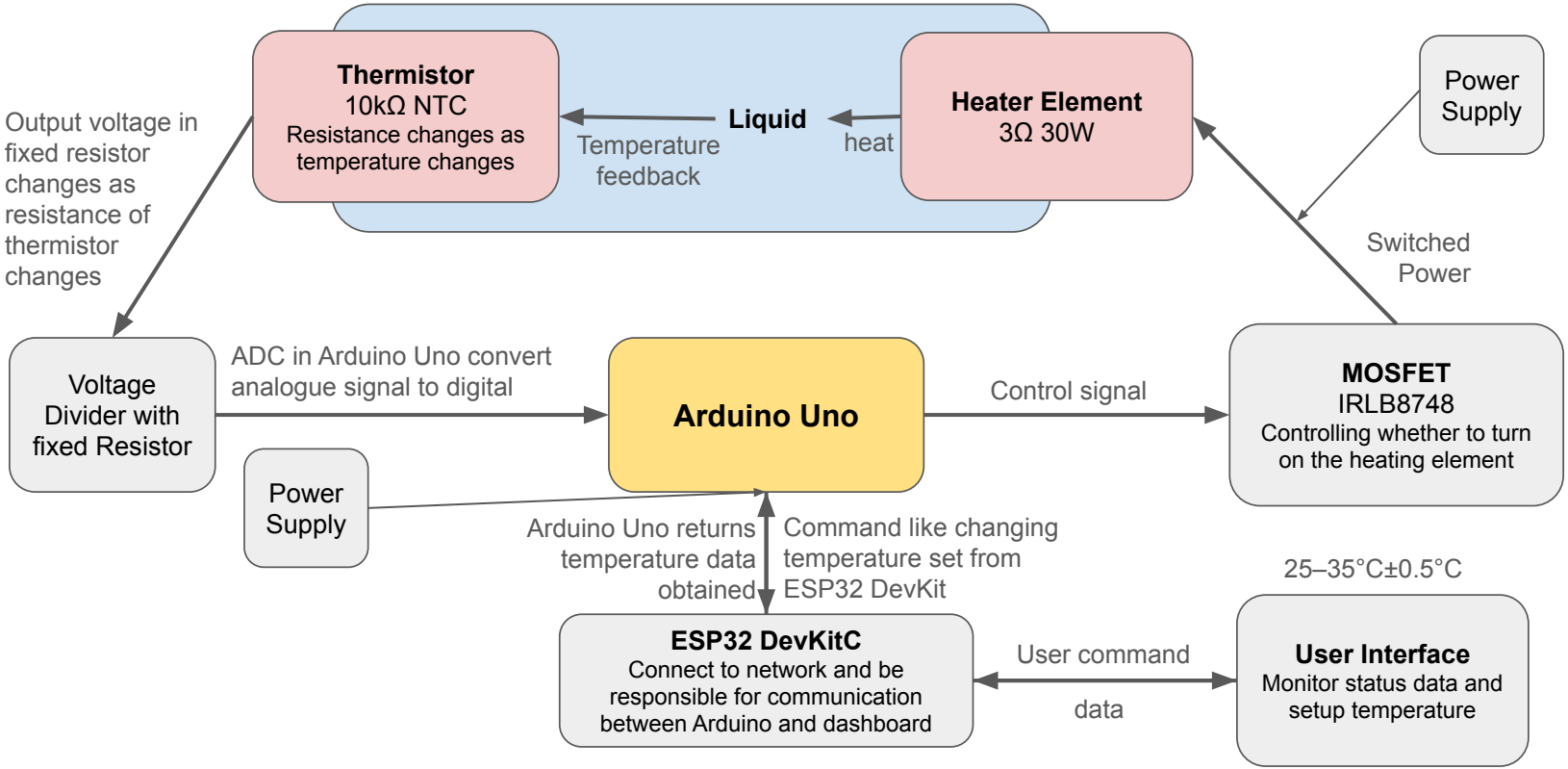


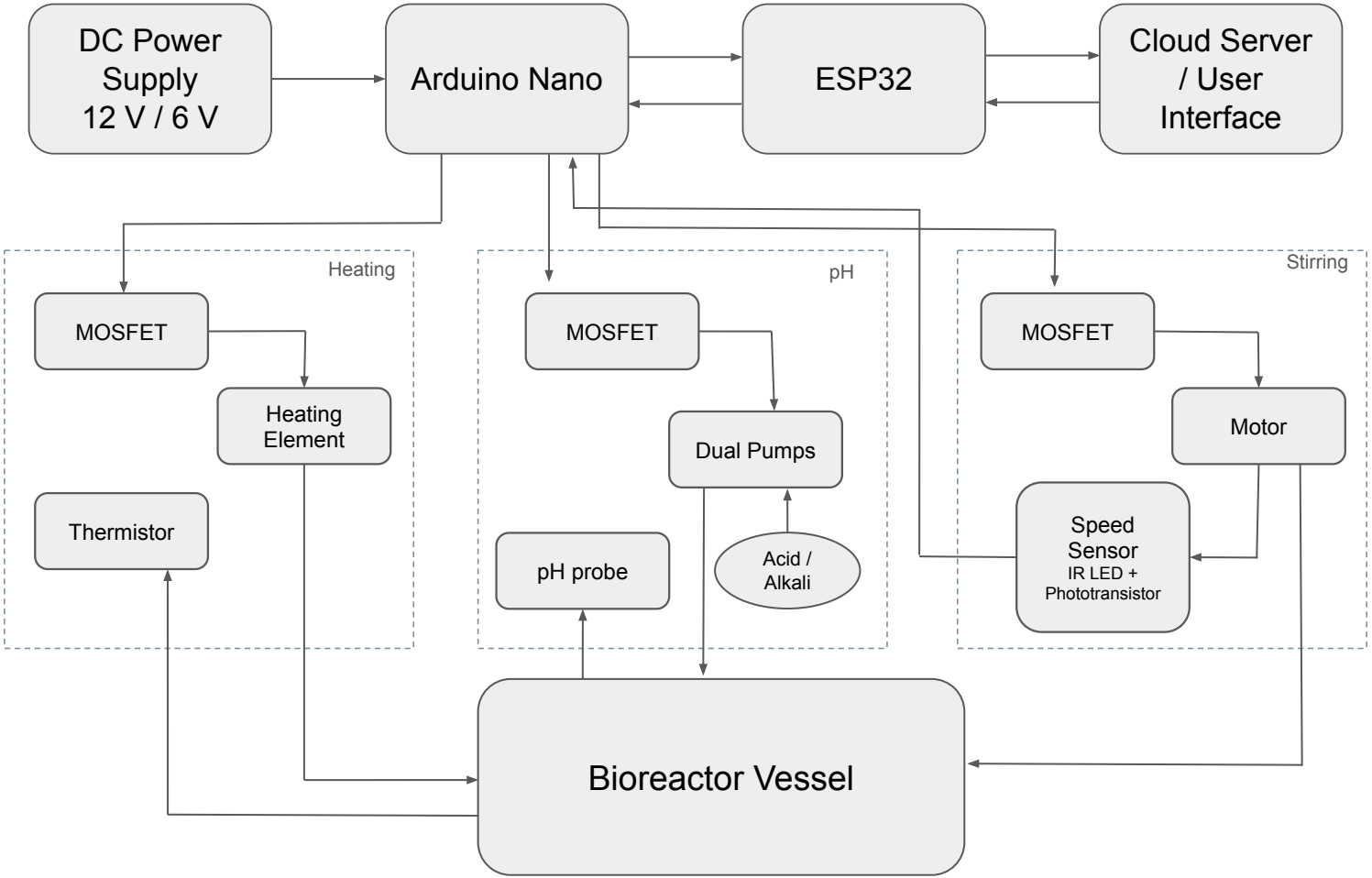
pH subsystem



By: abie

Heating subsystem




DC Power Supply 12 V / 6 V
Arduino Nano
ESP32
Cloud Server / User Interface
Heating
MOSFET
Heating Element
Thermistor
pH
MOSFET
Dual Pumps
pH probe
Acid / Alkali
Stirring
MOSFET
Motor
Speed Sensor
IR LED + Phototransistor
Bioreactor Vessel

Connectivity and User Interface

Shared monitoring, target adjustment and experiment logging

AT THE REACTOR | **NETWORK SERVICE** | **ON THE OPERATOR LAPTOP**

Control board
Sensors + actuators
Local feedback loops

UART ⟷

ESP32 gateway
Forwards readings
and target requests

Wi-Fi MQTT ⟷

MQTT broker
Routes messages
between clients

MQTT ⟷

Laptop backend
Hosts UI + CSV logs
Feeds data analysis

Local web ⟷

Browser
Dashboard
Operator inputs

Readings + controller confirmations → ← Target requests

PROPOSED SCREEN / example values

Temperature
29.8 °C
Confirmed: 30.0 °C
New target | Apply



25–35 °C

Stirring speed
998 RPM
Confirmed: 1000 RPM
New target | Apply



500–1500 RPM

pH
5.02
Confirmed: 5.00
New target | Apply



pH 3–7

OPERATOR ACTIONS

Monitor values and trends.

Apply validated setpoints.

Wait for controller confirmation.

Export timestamped records.

On link loss: mark readings stale.
Local loops follow the agreed policy.

PROPOSED TESTS 1 Hz reporting | 95% of confirmations ≤2 s | stale warning after 5 s

Proposed architecture. Verify board models and course-broker access. Brief: Design Specification pp. 4, 13, 17–18; Connectivity pp. 18–19.

Data Analysis

- Using hysteresis two thresholds
- Calculate median and MAD for modified z-score
- Thresholds decided using anomalous data

